Flood Mapping & Geospatial AI

Repository Investigation Report

Sentinel-1/2 | Kuro Siwo | Sen1Floods11 | SAR Change Detection | Infrastructure Impact | OSM Road Accessibility | Copernicus DEM | Geospatial AI Dashboards

Generated: 2026-10-10 | Investigator: Antigravity AI | 9 Repositories Investigated | 7 Topic Areas Covered

#1 ImpactMesh

IBM/ImpactMesh

Only repo with S1+S2+DEM+EMS labels.

Apache-2.0. pip-installable.

#2 KuroSiwo

Orion-AI-Lab/KuroSiwo

Best SAR segmentation code + weights.

MIT. Actively maintained 2026.

#3 Prithvi EO-2.0

NASA-IMPACT/hls-foundation-os

SOTA optical foundation model.

Apache-2.0. HuggingFace demo live.

! Key Gap

No End-to-End System Exists

Grounded reporting, OSM overlay,

dashboard: absent from ALL repos.

Flood Mapping & Geospatial AI - Repository Investigation Report

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Executive Summary

No single public repository covers the full hackathon brief end-to-end. The strongest individual-component baselines are Kuro Siwo (SAR flood segmentation) and IBM/ImpactMesh + Prithvi (multimodal, DEM-aware, foundation-model ready). The closest integrated competitor to the full brief is a manually assembled pipeline combining several of the repositories below - no single team has shipped all seven components in one open codebase.

1. Comprehensive Comparison Table

Tier 1 - Core SAR / Flood Segmentation Baselines

Attribute	Kuro Siwo	Sen1Floods11	ETCI-2021
URL	Orion-AI-Lab/KuroSiwo	cloudtostreet/Sen1Floods11	sidgan/ETCI-2021-Competition-on-Flood-Detection
Last code push	2026-07-16 Active	2021-04-09 Stale	2022-06-19 Stale
Stars / Forks	87 / 17	258 / 60	182 / 39
Open issues	9	13 (unanswered)	1
License	MIT	None stated	Apache-2.0
Language	Python	Jupyter Notebook	Jupyter Notebook
Dataset implemented	43 flood events, GRD + SLC, multi-temporal; v2 updated 2024	4,831 chips (512x512), 11 events, 6 continents; S1 GRD + S2 hand labels	ETCI 2021 competition data (Sentinel-1 GRD, global)
Data acquisition	Shell script + Dropbox + HuggingFace GeoTIFFs + webdatasets	gsutil rsync gs://sen1floods11 (~14 GB)	External download from competition host
Preprocessing	Yes - normalization constants, webdataset format	Yes - COG-spec GeoTIFFs, STAC catalog	Partial - normalization in notebooks
Training script	Yes - full training loop with configs	Jupyter notebooks only (Train.ipynb)	Jupyter notebooks + scripts; cyclical pseudo-labeling
Inference script	Yes - BlackBench evaluation suite	No standalone inference script	Inference notebook present
Model architectures	U-Net, ResNet-50/101, ViT variants via config	FCN, LogReg, Random Forest (comparison)	U-Net with semi-supervised extensions
Pretrained weights	Yes - HuggingFace checkpoints	Yes - GCS bucket pretrained_models/	No public weights
SAR change detection	YES - multi-temporal pre/co-event pairs	NO - uni-temporal only	NO - uni-temporal only

Attribute	Kuro Siwo	Sen1Floods11	ETCI-2021
Sentinel-2 fusion	NO - SAR only	YES - labels derived from S2	NO - SAR only
Copernicus DEM	NO	NO	NO
OSM road accessibility	NO	NO	NO
Infrastructure impact	NO	NO	NO
Geospatial dashboard	NO	NO	NO
Grounded reporting	NO	NO	NO
Geographic assumptions	Global; no hardcoded region; VV+VH polarization assumed	Global; 6 continents; 512x512 px chips	Global; chip-based; no tiling logic
Evaluation data	BlackBench held-out test; IoU/F1/precision/recall	Separate hand-labeled test set; IoU in paper	Competition leaderboard; F1 reported in paper
Reproducibility	HIGH - config files, seeds, eval scripts present	MEDIUM - data accessible; training code minimal	MEDIUM - notebooks work but competition-env dependent
Key dependencies	PyTorch, torchvision, wandb, webdataset, gdal	TensorFlow/Keras (older), gsutil	PyTorch, segmentation-models-pytorch, albumentations
Hardware requirements	GPU recommended (A100 tested; V100 feasible)	CPU for experiments; GPU for training	GPU (V100; ~12h training)
Setup difficulty	MEDIUM - clear README, conda env	HARD - GCS download + notebook-first	MEDIUM - requirements.txt available

Tier 2 - Multimodal and Foundation Model Repos

Attribute	IBM/ImpactMesh	NASA-IMPACT/hls-foundation-os (Prithvi)	UrbanSARFloods
URL	IBM/ImpactMesh	NASA-IMPACT/hls-foundation-os	jie666-6/UrbanSARFloods
Last code push	2026-09-25 Very active	2025-07-18 Active	2026-01-06 Active
Stars / Forks	15 / 3	407 / 97	41 / 3
Open issues	0	34	4
License	Apache-2.0	Apache-2.0	NONE STATED
Language	Python	Jupyter Notebook	Dataset only
Dataset	400+ events; S1 + S2 + CopDEM; 4 temporal obs; Copernicus EMS labels	Sen1Floods11 (fine-tuning); HLS multi-spectral (pre-training)	8,879 chips; 18 events; SLC intensity + InSAR coherence
Data acquisition	hf download ibm-esa-geospatial/ImpactMesh-Flood; Zarr v2	GCS bucket + NASA EarthData	HuggingFace Dataset4EO earthflow/UrbanSARFlood
Preprocessing	YES - Zarr dataloaders, TerraTorch normalization configs	PARTIAL - pre-tiled chips; normalization in config	MINIMAL - dataset README only
Training script	YES - terratorch fit --config configs/...	YES - MMSegmentation YAML configs	NO
Inference script	YES - terratorch test	YES - notebook + HuggingFace Spaces demo	NO
Model architectures	TerraMind ViT; any TerraTorch encoder	Prithvi ViT-MAE 100M/300M; UperNet/FPN decoder	U-Net, DeepLabV3+ (via SMP; not provided)
Pretrained weights	Via TerraTorch (auto-download)	HuggingFace: ibm-nasa-geospatial/Prithvi-EO-2.0-300M-TL-Sen1Floods11	NO weights
SAR change detection	YES - 4-timestamp multi-temporal	NO - optical only	YES - InSAR coherence
Sentinel-2 fusion	YES - S1 + S2 + DEM fused	YES - optical only (HLS = Landsat + S2)	NO - SAR only
Copernicus DEM	YES - 4th modality in training	NO	NO
OSM road accessibility	NO	NO	NO

Attribute	IBM/ImpactMesh	NASA-IMPACT/hls-foundation-os (Prithvi)	UrbanSARFloods
Infrastructure impact	PARTIAL - Copernicus EMS building damage labels	NO	NO
Geospatial dashboard	NO	HuggingFace Spaces demo only	NO
Grounded reporting	NO	NO	NO
Geographic assumptions	Global; no hardcoded bbox; DEM resolution varies by tile	HLS UTM projection; 224x224 px hardcoded	Global; 5 continents; 512x512 chips
Hardcoded params	num_classes: 3 (flood), 2 (fire); Zarr chunk sizes	6-band input (B02,B03,B04,B8A,B11,B12); 224x224 px mandatory	None (dataset only)
Evaluation data	HuggingFace test split; terratorch test gives IoU/F1	Sen1Floods11 test set; IoU in HuggingFace model card	Paper benchmark split; no evaluation script
Reproducibility	HIGH - pip installable, config-driven	MEDIUM - 34 open issues; env setup painful	LOW - no training code
Key dependencies	terratorch==1.2.1, zarr==2.18.0, numcodecs==0.15.1	mmsegmentation, mmcv, torch (pinned older versions)	N/A
Hardware requirements	GPU; >=16GB VRAM for TerraMind fine-tuning	A100 used; V100 feasible	N/A
Setup difficulty	LOW - pip install works	HARD - mmseg version conflicts in issues	LOW (data only)

Tier 3 - Operational and Domain-Specific Tools

Attribute	Sentinel1-Flood-Finder	S1GFlood-Detection (DAM-Net)	OSMnx
URL	cordmaur/Sentinel1-Flood-Finder	Tamer-Saleh/S1GFlood-Detection	gboeing/osmnx
Last code push	2024-07-15 Slowing	UNKNOWN (README 404 at investigation)	Actively maintained
Stars / Forks	38 / 5	Unknown	~5k / ~800
License	GPL-3.0 COPYLEFT	Unknown	MIT
Language	Python (Jupyter-heavy)	Python	Python
Core capability	Random Forest flood detection on Planetary Computer; no deep learning	ViT bi-temporal change detection (DAM-Net); 5,360 S1 pairs	OSM street network download, graph analysis, routing
Data acquisition	MS Planetary Computer API (auto)	GCS or direct download	Overpass API (auto)
Preprocessing	Built-in (speckle filter, calibration)	Paper-described; code not confirmed public	N/A (vector data)
Training	No training; pre-fitted RF	Paper-described; code availability unclear	N/A
Inference	Yes - one-call flood map generation	Notebook-described	N/A
SAR change detection	NO - single-image threshold	YES - bi-temporal ViT	N/A
DEM integration	NO	NO	N/A
OSM road accessibility	NO	NO	YES - full graph analysis, isochrones, shortest path
Flood x OSM overlay	NO	NO	NO (requires user integration)
Disconnected settlements	NO	NO	YES with user implementation
Dashboard	NO	NO	NO (folium/matplotlib visualization only)
Geographic assumptions	Planetary Computer coverage only; no global offline	Global S1 GRD pairs	Overpass-dependent; offline via .osm.pbf
Hardcoded params	RF threshold values; PC API keys required	Threshold in ViT head	bbox per-call; graph simplification defaults
Evaluation	Qualitative only	BlackBench / competition	Graph metrics via networkx
Reproducibility	MEDIUM - API keys required	LOW - README 404	HIGH - well-documented stable API

Attribute	Sentinel1-Flood-Finder	S1GFlood-Detection (DAM-Net)	OSMnx
Key dependencies	planetary_computer, pystac_client, rioxarray	PyTorch, einops, timm	networkx, shapely, geopandas, requests
Hardware	CPU (RF model)	GPU recommended	CPU
Setup difficulty	MEDIUM - PC auth needed	UNKNOWN	LOW

2. Coverage Matrix vs. Hackathon Brief

Component	Best Available Coverage	Critical Gap
Sentinel-1 SAR flood segmentation	Kuro Siwo (multi-temporal, GRD+SLC)	No urban coherence by default
Sentinel-2 optical fusion	ImpactMesh (S1+S2+DEM fused)	Fusion at dataset level; no runtime S2 fetch
Kuro Siwo dataset	Kuro Siwo repo itself	Generalization to new AOIs needs fine-tuning
Sen1Floods11 benchmark	Sen1Floods11 + Prithvi	Training code minimal in official repo
SAR change detection	ImpactMesh (4-temporal), UrbanSARFloods (coherence)	No single end-to-end change detection pipeline exists
Deep learning segmentation	Prithvi EO-2.0 / KuroSiwo / ETCI-2021	All chip-based; no tile-stitching for full-scene inference
Infrastructure impact	ImpactMesh (Copernicus EMS annotations)	No building-level damage report generation
Copernicus DEM terrain analysis	ImpactMesh (DEM as 4th modality)	No depth estimation or slope analysis module
OSM road accessibility	OSMnx	No flood-layer integration; must be custom-built
Disconnected settlements	OSMnx (graph fragmentation)	No automated settlement identification
Geospatial AI dashboard	HuggingFace Spaces (Prithvi demo only)	No operational emergency-responder dashboard
Grounded incident reporting	NONE FOUND	Entire component absent from all investigated repos

3. Ranking by Relevance to Hackathon Brief

Rank 1 - IBM/ImpactMesh

Score: 7/7 topic areas partially addressed

- Only repo that simultaneously integrates S1 SAR, S2 optical, Copernicus DEM, multi-temporal change detection, and Copernicus EMS ground truth annotations.
- `pip install impactmesh` works. TerraTorch training config is clean and reproducible.
- Weaknesses: No OSM overlay, no dashboard, no road accessibility, no reporting layer. Created Nov 2025 - very new, low adoption (15 stars), arXiv paper listed as "coming soon" at investigation time.

Rank 2 - Orion-AI-Lab/KuroSiwo

Score: 4/7 topic areas covered

- Highest-quality SAR-only flood segmentation dataset and training code in the field.
- Actively maintained (last push July 2026). MIT license. Pretrained weights downloadable.

- BlackBench provides a rigorous, standardized evaluation protocol.
- Weaknesses: No optical fusion, no DEM, no infrastructure layer, no dashboard. SAR only.

Rank 3 - NASA-IMPACT/hls-foundation-os (Prithvi EO-2.0)

Score: 3/7 topic areas (optical-centric)

- Best foundation model for Sentinel-2 flood segmentation; strong transfer learning across geographies.
- 407 stars, 97 forks. Apache-2.0. HuggingFace Spaces demo publicly accessible today.
- Weaknesses: No SAR input; 6-band HLS hardcoded; 34 open issues; complex MMSeg environment setup.

Rank 4 - cloudtostreet/Sen1Floods11

Score: 3/7 topic areas

- Canonical benchmark dataset; cited in virtually every competing paper.
- GCS bucket access stable; STAC catalog present for geospatial tooling.
- Weaknesses: No license declared on GitHub; code is notebook-only; last push 2021; 13 open issues with no responses.

Rank 5 - sidgan/ETCI-2021-Competition-on-Flood-Detection

Score: 2/7 topic areas

- Best example of a competition-winning SAR segmentation pipeline with semi-supervised learning.
- Operationally deployed by Orange Business Services - inference achieves 1.22 seconds per 63,000 km² tile.
- Apache-2.0. Documented presentations, poster, and paper available.
- Weaknesses: No multimodal fusion, no terrain, no OSM, no dashboard. Code not updated since June 2022.

Rank 6 - jie666-6/UrbanSARFloods

Score: 2/7 topic areas

- Unique value: SLC coherence for urban flood detection - addresses the hardest SAR scenario (double-bounce cancellation in cities).
- 2024 paper; actively maintained through early 2026.
- Weaknesses: Dataset-only repository; no training code; no license stated.

Rank 7 - gboeing/osmnx

Score: 2/7 topic areas

- The only reliable, production-quality library for OSM road network analysis and graph computations.
- MIT license; excellent documentation; stable API; no setup issues.
- Weaknesses: Not a flood-specific tool; flood overlay and disconnected settlement detection are entirely custom implementation burden.

Rank 8 - cordmaur/Sentinel1-Flood-Finder

Score: 1/7 topic areas

- Fast, cloud-native Random Forest approach; no GPU required; beginner-friendly.
- Weaknesses: GPL-3.0 copyleft risk; no deep learning; Planetary Computer API lock-in; development slowing since July 2024.

Rank 9 - S1GFlood-Detection (DAM-Net)

Score: 1/7 topic areas

- Interesting bi-temporal ViT architecture with global SAR dataset.
- Weaknesses: README was 404 at investigation time; code availability unconfirmed; do not build dependencies on this.

4. Closest End-to-End Competitors

No public repository implements the complete hackathon brief. The closest approximation requires assembling multiple repos manually:

```
ImpactMesh (S1 + S2 + DEM + Copernicus EMS labels)
+ KuroSiwo pretrained weights (SAR backbone)
+ OSMnx (road graph with networkx)
+ Custom integration code:
flood mask GeoJSON --> geopandas spatial join --> edge removal --> connected components
+ Custom dashboard (Streamlit + Folium/MapLibre)
+ Custom grounded reporting (LLM + GeoJSON-to-NL generation)
```

Academic near-competitors (partial code only):

- *Ambulance STARS* (MDPI 2026): S1 flood detection + OSM routing for ambulance accessibility. No public GitHub repository found.
- *OpenFloodRiskMap (OFRM)*: Web decision support for road flood risk. Appears proprietary/institutional.
- *Copernicus EMS INFLOS*: Operational DEM-based depth estimation. Not open-source.

5. Strongest Reusable Research Baselines

Use Case	Recommended Baseline	Rationale
SAR backbone weights	KuroSiwo pretrained ResNet/UNet on HuggingFace	MIT; multi-temporal; globally trained on 43 events
Multimodal training data	ImpactMesh-Flood on HuggingFace	Apache-2.0; S1+S2+DEM; 200+ flood events

Use Case	Recommended Baseline	Rationale
Optical foundation model	Prithvi-EO-2.0-300M-TL-Sen1Floods11	Apache-2.0; SOTA on Sen1Floods11; HuggingFace-native
Urban coherence detection	UrbanSARFloods dataset via Dataset4EO	Unique coherence channel; addresses urban SAR ambiguity
Road network graph engine	OSMnx	MIT; production-grade; networkx integration
Standard benchmark eval	Sen1Floods11 test set + KuroSiwo BlackBench	Community standards; enables cross-paper comparison
Semi-supervised training	ETCI-2021 pseudo-labeling code	Apache-2.0; proven at 63,000 km² scale

6. Component Action Plan

Independently Implementable (no license risk)

- OSM road accessibility + disconnected settlement detection: Build with `osmnx` (MIT) and `geopandas`. Overlay flood mask GeoJSON onto road graph; remove flooded edges; compute `networkx.connected_components`. No existing repo does this end-to-end - pure custom work estimated at ~2-3 days.
- Copernicus DEM terrain analysis: Use `rasterio` + `richdem` or `pysheds`. DEM tiles freely downloadable from AWS S3 (`s3://copernicus-dem-30m`). No standalone public module exists for this.
- Geospatial dashboard: Streamlit + Folium/Deck.gl. No direct competitor in the open-source space for this specific use case.
- Grounded incident reporting: LLM + structured geospatial output (GeoJSON to natural language narrative). Entirely novel component; no public repo found.

Adapt with Attribution (license permits; citation required)

- SAR flood segmentation model: Use KuroSiwo pretrained weights (MIT). Fine-tune on target AOI with custom data. Cite Bountos et al. 2024 (arXiv:2311.12056).
- Multimodal training pipeline: Fork ImpactMesh (Apache-2.0). Add custom inference entrypoint for new AOI. Cite IBM/DLR/ESA Phi-lab.
- Semi-supervised pseudo-labeling: Adapt ETCI-2021 cyclical pseudo-labeling code (Apache-2.0). Cite Paul and Ganju 2021 (arXiv:2107.08369).
- Foundation model fine-tuning: Use Prithvi-EO-2.0 via TerraTorch (Apache-2.0). Cite IBM NASA 2023 Prithvi paper.

Inspiration Only (license constraints or data restrictions)

- Sen1Floods11 data: No license declared on GitHub - commercial and competition use is legally ambiguous. Use only as an evaluation benchmark. Cite Bonafilia et al. 2020 (CVPRW 2020).
- Sentinel1-Flood-Finder: GPL-3.0 - if the hackathon submission must be Apache or proprietary, GPL is incompatible. Use as architectural reference only.
- UrbanSARFloods: No license. Do not redistribute the dataset or derivatives without explicit author permission.
- DAM-Net (S1GFlood): Code availability unverified (README 404). Do not build a dependency on this.
- Copernicus EMS annotations (via ImpactMesh): Copernicus open license permits research; verify non-commercial restrictions for hackathon prize eligibility before submitting.

7. Missing Components (Must Build From Scratch)

The following components are completely absent from all investigated repositories:

1. Real-time Sentinel-1 ingestion pipeline - CDSE API or Planetary Computer query for new event acquisitions
2. Automated pre/post-event pair selection - orbit matching, temporal window logic, scene metadata parsing
3. Flood mask to road network overlay - GeoDataFrame spatial join + edge removal from OSM graph
4. Disconnected settlement identification - graph component labeling + settlement polygon lookup from OSM or GHSL
5. DEM-informed flood depth estimation - flood water surface interpolation minus DEM bare-earth elevation
6. Grounded incident reporting - natural language generation from structured geospatial data (LLM-based)
7. Interactive emergency-responder dashboard - real-time map, alert feed, per-settlement accessibility score
8. Full-pipeline evaluation harness - end-to-end F1 at settlement level, not just chip-level pixel accuracy

8. Data Restriction Checklist for Hackathon Submission

Data Source	License	Hackathon-Safe	Notes
Sentinel-1 GRD (ESA Copernicus)	Copernicus Open License	YES	Free for commercial use with attribution
Sentinel-2 L2A (ESA Copernicus)	Copernicus Open License	YES	Same as S1
Copernicus DEM GLO-30	Open (TanDEM-X derived)	YES	CC Attribution required
OpenStreetMap	ODbL	YES	Share-alike for derived databases; verify if output maps qualify
Kuro Siwo dataset	MIT	YES	Citation required
ImpactMesh dataset	Apache-2.0	YES	Citation required
Sen1Floods11 dataset	UNLICENSED	UNCLEAR	Contact Cloud to Street before use in competition
UrbanSARFloods dataset	UNLICENSED	UNCLEAR	Contact authors (Jie Zhao, TUM)
Copernicus EMS products	Copernicus Open License	YES	Attribution required
HLS (NASA) data	NASA Open Data	YES	US Government works; public domain

This report is based on direct repository metadata retrieved via GitHub API, README file contents, commit history, and web research conducted on 2026-10-09. Code that was not publicly accessible (DAM-Net) has been explicitly flagged. Claims marked as verified were cross-checked against actual file contents or API responses. Claims marked unclear require further verification before relying on them for a competition submission.